NEET UG • HIGH-YIELD PHYSICS TUTORIAL

Magnetism and Matter

Complete NCERT Foundation, Mathematical Derivations, Conceptual Analysis, and High-Yield Practice Questions

1. Syllabus Overview & Learning Objectives

Magnetism is a fundamental branch of electromagnetism that describes magnetic dipole interactions, field characteristics, Gauss's law for magnetic fields, and the macroscopic magnetic classification of materials. In the NEET UG examination, questions from this chapter consistently test core conceptual distinctions, magnetic potential energy calculations, torque, dipole analog formulations, and quantitative relationships involving magnetization, susceptibility, and permeability.

Key NEET UG Learning Objectives:

- State the fundamental properties of magnetic field lines and explain why magnetic monopoles do not exist.
- Model a bar magnet as an equivalent current-carrying finite solenoid and deduce its axial and equatorial fields.
- Calculate the torque $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$ and magnetic potential energy $U = -\mathbf{m} \cdot \mathbf{B}$ in a uniform magnetic field.
- Apply Gauss's law for magnetism, $\oint \mathbf{B} \cdot d\mathbf{S} = 0$, to open and closed surfaces.
- Define and distinguish magnetic intensity ($\mathbf{H}$), magnetization ($\mathbf{M}$), magnetic susceptibility (χ), and relative permeability (μ_r).
- Compare the macroscopic and microscopic behaviors of diamagnetic, paramagnetic, and ferromagnetic materials.
- Understand the Meissner effect in superconductors and distinguish between soft and hard ferromagnets.

BOSONANCE

2. The Bar Magnet and Magnetic Field Lines

2.1 Universal Magnetic Phenomena

Historically originating from deposits of natural magnetic iron ore (magnetite or lodestone) discovered in the Magnesia district of Greece, magnetism is universal across macroscopic and microscopic scales. The fundamental observational laws of isolated bar magnets include:

- **Directive Property:** A freely suspended horizontal bar magnet aligns itself approximately along the geographic North-South direction. The end pointing toward the geographic north is designated as the *magnetic north pole* (N), and the opposite end is the *magnetic south pole* (S).
- **Attractive and Repulsive Forces:** Like magnetic poles repel each other, whereas unlike magnetic poles attract each other with an inverse-square force at sufficient separation.
- **Non-existence of Magnetic Monopoles:** Breaking a bar magnet transverse or parallel to its length yields two smaller magnets, each possessing an indivisible pair of North and South poles. Isolated magnetic monopoles do not exist in nature.

2.2 Properties of Magnetic Field Lines

Magnetic field lines provide a continuous visual map of the magnetic induction vector $\mathbf{B}$. Their defining physical characteristics are:

1. **Continuous Closed Loops:** Unlike electrostatic field lines—which originate on positive electric charges and terminate on negative electric charges—magnetic field lines form unbroken closed loops. Outside the magnet, they

run from the North pole to the South pole; inside the magnetic material, they run continuously from the South pole to the North pole.

2. **Direction of Field:** The tangent drawn to a magnetic field line at any point indicates the direction of the net magnetic field vector $\mathbf{B}$ at that point.
3. **Line Density and Field Strength:** The spatial density of field lines (number of lines passing perpendicularly through a unit cross-sectional area) is directly proportional to the magnitude of $\mathbf{B}$. Denser lines indicate a stronger field.
4. **No Intersection:** Two magnetic field lines can never cross. If they intersected, two distinct tangents could be drawn at the intersection point, implying two different directions of the net magnetic field at the same point, which is physically impossible.

> **NEET Precision Note:** In older texts, magnetic field lines were termed "magnetic lines of force." This nomenclature is strictly avoided in modern physics because, unlike electrostatics where $\mathbf{F}_e = q\mathbf{E}$ points along $\mathbf{E}$, the magnetic Lorentz force $\mathbf{F}_m = q(\mathbf{v} \times \mathbf{B})$ is perpendicular to the magnetic field vector $\mathbf{B}$. Thus, field lines do not indicate the direction of magnetic force on a moving charge.

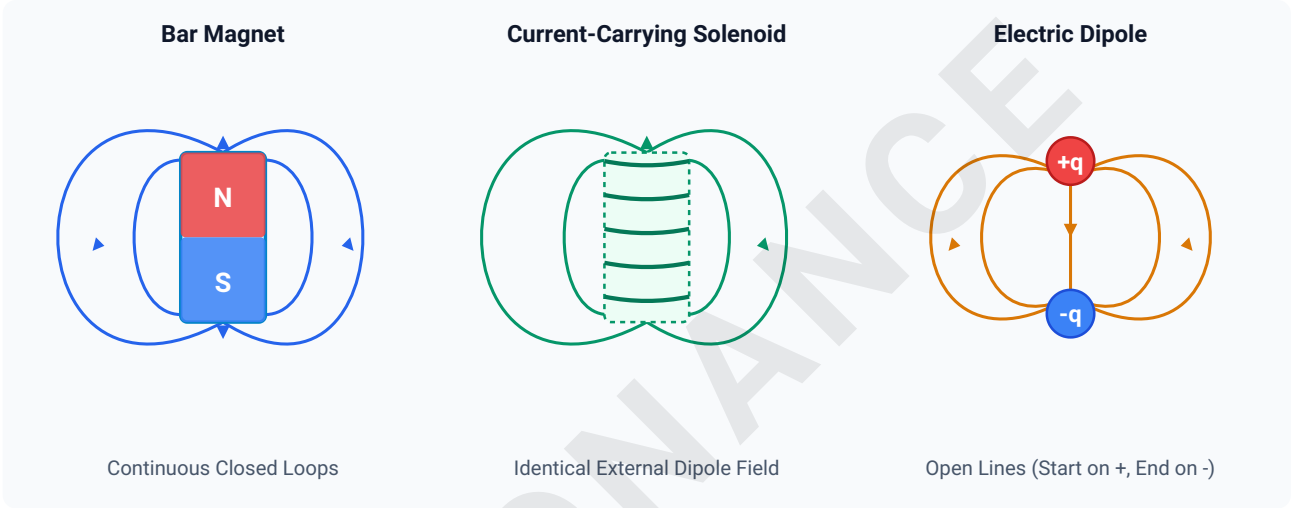


Figure 1: Comparison of field lines for a bar magnet, a current-carrying solenoid, and an electric dipole. Note that magnetic lines form continuous closed loops through the interior, while electric lines terminate on point charges.

BOSONANCE

3. Bar Magnet as an Equivalent Finite Solenoid

According to **Ampère's hypothesis**, all magnetic phenomena in bulk materials can be explained in terms of circulating microscopic atomic currents. Slicing a bar magnet in half cuts these elementary current loops into two sets of circulating currents, forming two complete, smaller bar magnets.

To mathematically demonstrate the equivalence between a bar magnet and a finite solenoid, consider a cylindrical solenoid of length $2l$, radius a, carrying current I, and having n turns per unit length. Let P be an axial observation point located at distance r from the center O of the solenoid, where $r \gg a$ and $r \gg l$.

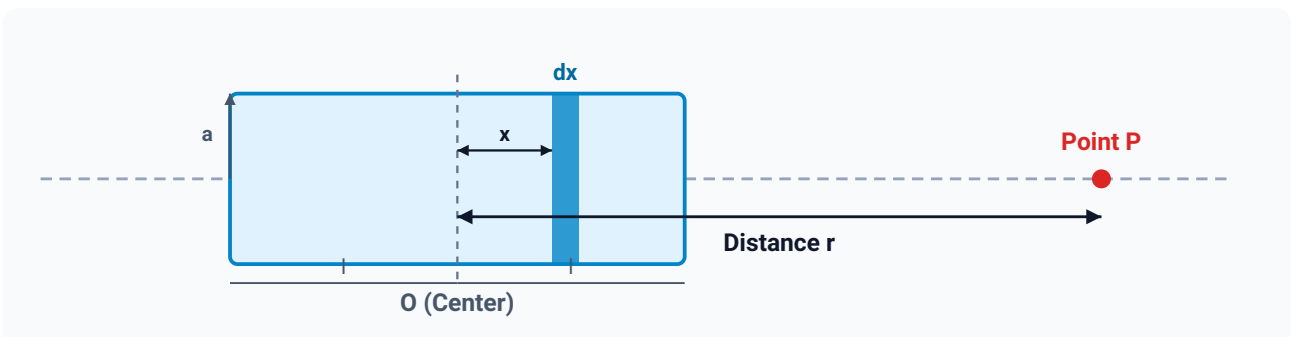


Figure 2: Setup for computing the magnetic field along the axial line of a finite solenoid of length $2l$ and radius a.

Consider an elementary circular slice of width dx at a displacement x from the center O. The number of turns in this element is $n\,dx$. Using the formula for the axial magnetic field due to a single circular current loop of radius a at distance $(r - x)$:

$$dB = \frac{\mu_0 (n\,dx) I a^2}{2\left[(r-x)^2 + a^2\right]^{3/2}}$$

Under the far-field approximation where $r \gg a$ and $r \gg x$ (since $|x| \leq l \ll r$), we can approximate the denominator as:

$$\left[(r-x)^2 + a^2\right]^{3/2} \approx (r^2)^{3/2} = r^3$$

Substituting this approximation into the expression for dB and integrating from $x = -l$ to $x = +l$:

$$B = \frac{\mu_0 n I a^2}{2r^3} \int_{-l}^{l} dx = \frac{\mu_0 n I a^2}{2r^3}[2l] = \frac{\mu_0 I (2l \cdot n) a^2}{2r^3}$$

The total number of turns in the solenoid is $N = 2l \cdot n$. Multiplying and dividing by π:

$$B = \frac{\mu_0 I N (\pi a^2)}{2\pi r^3}$$

Recognizing that $A = \pi a^2$ is the cross-sectional area of each turn, the total magnetic dipole moment of the solenoid is $m = NIA = (2l \cdot n) I (\pi a^2)$. Rearranging into standard form:

$$\mathbf{B}_{\text{axial}} = \frac{\mu_0}{4\pi} \frac{2\mathbf{m}}{r^3} \quad (\text{for } r \gg l)$$

This equation is mathematically identical to the axial field of a short bar magnet. Hence, a bar magnet of magnetic moment $\mathbf{m}$ produces a magnetic field entirely equivalent to a solenoid of identical dipole moment.

BOSONANCE

4. Magnetic Dipole in a Uniform Magnetic Field

4.1 Torque Formulation

When a magnetic dipole of magnetic moment $\mathbf{m}$ (such as a compass needle or bar magnet) is placed in a uniform external magnetic field $\mathbf{B}$ at an angle θ:

- **Net Translational Force:** Since the field is uniform, the effective forces on the two poles are equal in magnitude ($q_m B$) and opposite in direction. Thus, $\mathbf{F}_{\text{net}} = 0$.
- **Restoring Torque:** The two forces act along different lines of action separated by a perpendicular lever arm, producing a net torque:

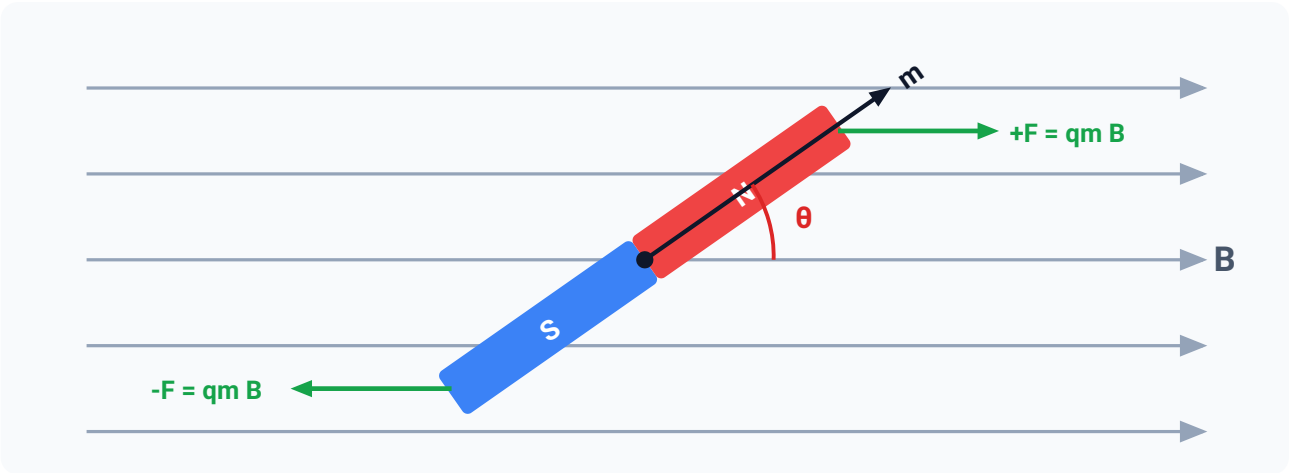


Figure 3: Magnetic needle with magnetic moment $\mathbf{m}$ aligned at an angle θ to a uniform magnetic field $\mathbf{B}$, experiencing an aligned couple.

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B} \quad \Longrightarrow \quad \tau = mB\sin\theta$$

The torque acts to align the magnetic moment vector $\mathbf{m}$ with the field vector $\mathbf{B}$.

4.2 Magnetic Potential Energy and Equilibrium States

The work done by an external agent against the magnetic restoring torque to rotate the dipole through an angle $d\theta$ is $dW = \tau_{\text{ext}}\, d\theta = mB\sin\theta\, d\theta$. Choosing the reference zero potential energy orientation at $\theta_0 = 90^\circ$ ($\pi/2$ radians):

$$U_m = \int_{90^\circ}^{\theta} mB\sin\theta\, d\theta = -mB\cos\theta = -\mathbf{m}\cdot\mathbf{B}$$

This potential energy relation yields two equilibrium configurations:

Orientation (θ)	Torque (τ)	Potential Energy (U_m)	Equilibrium Classification
$\theta = 0^\circ$ ($\mathbf{m} \parallel \mathbf{B}$)	$\tau = 0$	$U_{\min} = -mB$	**Stable Equilibrium** (Minimum energy; small angular displacements produce restoring torque)
$\theta = 90^\circ$ ($\mathbf{m} \perp \mathbf{B}$)	$\tau_{\max} = mB$	$U = 0$	Non-equilibrium (Maximum torque acting on dipole)
$\theta = 180^\circ$ ($\mathbf{m}$ antiparallel to $\mathbf{B}$)	$\tau = 0$	$U_{\max} = +mB$	**Unstable Equilibrium** (Maximum energy; small angular displacements accelerate dipole away)

The work required to rotate the magnet from initial orientation θ_1 to final orientation θ_2 is:

$$W = \Delta U = U(\theta_2) - U(\theta_1) = -mB(\cos\theta_2 - \cos\theta_1) = mB(\cos\theta_1 - \cos\theta_2)$$

5. The Electrostatic Analog

The mathematical formalisms governing short magnetic dipoles at large distances ($r \gg l$) map directly onto the relations for electric dipoles through the classical replacement dictionary:

Physical Quantity	Electrostatics	Magnetism
Permittivity / Permeability Constant	$\dfrac{1}{\varepsilon_0}$	μ_0
Coupling Multiplier	$\dfrac{1}{4\pi\varepsilon_0}$	$\dfrac{\mu_0}{4\pi} = 10^{-7}\,\mathrm{T\cdot m\cdot A^{-1}}$
Dipole Moment	Electric dipole moment $\mathbf{p}$	Magnetic dipole moment $\mathbf{m}$
Field Vector	Electric field $\mathbf{E}$	Magnetic induction $\mathbf{B}$
Axial Field ($r \gg l$)	$\mathbf{E}_A = \dfrac{1}{4\pi\varepsilon_0}\dfrac{2\mathbf{p}}{r^3}$	$\mathbf{B}_A = \dfrac{\mu_0}{4\pi}\dfrac{2\mathbf{m}}{r^3}$
Equatorial Field ($r \gg l$)	$\mathbf{E}_E = -\dfrac{1}{4\pi\varepsilon_0}\dfrac{\mathbf{p}}{r^3}$	$\mathbf{B}_E = -\dfrac{\mu_0}{4\pi}\dfrac{\mathbf{m}}{r^3}$
Ratio of Axial to Equatorial Magnitude	$\dfrac{E_A}{E_E} = 2$	$\dfrac{B_A}{B_E} = 2$
External Torque	$\boldsymbol{\tau} = \mathbf{p}\times\mathbf{E}$	$\boldsymbol{\tau} = \mathbf{m}\times\mathbf{B}$
Potential Energy	$U = -\mathbf{p}\cdot\mathbf{E}$	$U = -\mathbf{m}\cdot\mathbf{B}$

BOSONANCE

6. Gauss's Law for Magnetism

6.1 Integral Statement and Physical Significance

In electrostatics, Gauss's law establishes that the net electric flux through any arbitrary closed Gaussian surface S is proportional to the net enclosed electric charge: $\oint_S \mathbf{E} \cdot d\mathbf{S} = q_{\text{enc}}/\varepsilon_0$. If a surface encloses an isolated positive charge, field lines diverge outward, giving a positive net flux.

Because magnetic field lines form unbroken continuous loops, every magnetic field line entering any closed volume must inevitably exit that same volume. There are no sources (divergence points) or sinks (convergence points) for $\mathbf{B}$.

$$\Phi_B = \oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Fundamental Physics Conclusion: Gauss's law for magnetism confirms the empirical fact that *isolated magnetic poles (monopoles) do not exist*. The most fundamental magnetic entity is a dipole or current loop.

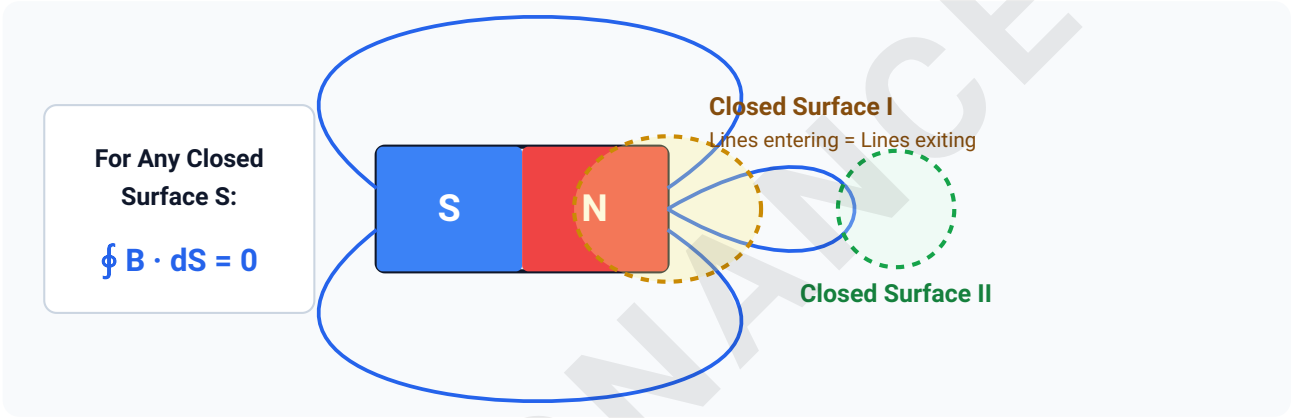


Figure 4: Gauss's Law in magnetism. Any closed Gaussian surface enclosing part or all of a magnet has zero net magnetic flux because field lines are continuous closed loops.

Hypothetical Existence of Monopoles: If isolated magnetic charges (magnetic monopoles with strength q_m) existed, Gauss's law for magnetism would be formally modified to:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \mu_0 q_{m,\text{enc}}$$

where $q_{m,\text{enc}}$ represents the net magnetic charge enclosed by surface S.

7. Magnetization, Magnetic Intensity, and Permeability

When materials are subjected to an external magnetic field, their atomic constituents respond to modify the internal magnetic environment. To quantitatively analyze these phenomena, four primary vectors and constants are defined:

7.1 Magnetization Vector ($\mathbf{M}$)

Electrons possess intrinsic spin and orbital angular momentum, each behaving as a tiny circulating current loop with an associated magnetic dipole moment. The **magnetization** $\mathbf{M}$ of a material is defined as the net vector magnetic moment per unit volume of the substance:

$$\mathbf{M} = \frac{\mathbf{m}_{\text{net}}}{V}$$

BOSONANCE

The SI unit of magnetization is ampere per meter ($\mathrm{A \cdot m^{-1}}$), and its dimensional formula is $[\mathrm{L^{-1}A}]$.

7.2 Magnetic Field Inside a Solenoid with Core

Consider a long solenoid having n turns per unit length carrying current I. In vacuum, the interior uniform magnetic field is:

$$\mathbf{B}_0 = \mu_0 n I$$

When the core of the solenoid is filled with a magnetic medium, atomic dipoles align, creating an additional magnetic field contribution $\mathbf{B}_m$ proportional to the material's magnetization:

$$\mathbf{B}_m = \mu_0 \mathbf{M}$$

The total magnetic induction $\mathbf{B}$ inside the core is the vector sum:

$$\mathbf{B} = \mathbf{B}_0 + \mathbf{B}_m = \mu_0 n I + \mu_0 \mathbf{M}$$

7.3 Magnetic Intensity Vector ($\mathbf{H}$)

To isolate the contribution of external free conduction currents from the internal bound atomic response, the **magnetic intensity** (or magnetizing field) $\mathbf{H}$ is defined as:

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} \quad \iff \quad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$$

Inside a long solenoid, the field purely due to free conduction current is $H = nI$. The SI unit of $\mathbf{H}$ is also ampere per meter ($\mathrm{A \cdot m^{-1}}$), having dimensions $[\mathrm{L^{-1}A}]$.

7.4 Magnetic Susceptibility (χ) and Permeability (μ_r, μ)

For linear and isotropic magnetic media, the induced magnetization is directly proportional to the applied magnetizing field:

$$\mathbf{M} = \chi \mathbf{H}$$

where χ (chi) is a dimensionless scalar known as the **magnetic susceptibility**. Substituting $\mathbf{M} = \chi\mathbf{H}$ into the total field equation:

$$\mathbf{B} = \mu_0(\mathbf{H} + \chi\mathbf{H}) = \mu_0(1+\chi)\mathbf{H} = \mu_0\mu_r\mathbf{H} = \mu\mathbf{H}$$

The central inter-relationships are:

$$\mu_r = 1 + \chi, \qquad \mu = \mu_0\mu_r = \mu_0(1+\chi)$$

- μ_r: **Relative magnetic permeability** (dimensionless ratio μ/μ_0, equivalent to dielectric constant K in electrostatics).
- μ: **Absolute magnetic permeability** of the medium, measured in $\mathrm{T \cdot m \cdot A^{-1}}$ or $\mathrm{N \cdot A^{-2}}$.

8. Classification of Magnetic Materials

Bulk materials are categorized into three major classes based on their atomic structure and magnetic susceptibility χ:

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Atomic Dipole Moment	Zero resultant dipole moment per atom in absence of external field.	Each individual atom possesses a permanent magnetic dipole moment.	Permanent atomic moments spontaneously aligned in cooperative domains.
Action in Non-Uniform Field	Repelled weakly; tends to move from *stronger to weaker* field regions.	Attracted weakly; tends to move from *weaker to stronger* field regions.	Attracted strongly; moves vigorously from *weaker to stronger* field regions.

BOSONANCE

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Susceptibility (χ)	Small and negative ($-1 \leq \chi < 0$), order of -10^{-5}.	Small and positive ($0 < \chi < \varepsilon$), order of $+10^{-5}$.	Large and positive ($\chi \gg 1$), typically 10^2 to 10^5.
Relative Permeability (μ_r)	Slightly less than unity ($0 \leq \mu_r < 1$).	Slightly greater than unity ($1 < \mu_r < 1 + \varepsilon$).	Much greater than unity ($\mu_r \gg 1$, often > 1000).
Temperature Dependence	Nearly independent of temperature.	Inversely proportional to absolute temperature: $\chi \propto 1/T$ (Curie's Law).	Decreases with temperature; transitions to paramagnetism at high temperature.
Representative Materials	Bismuth, Copper, Lead, Silicon, Nitrogen (at STP), Water, NaCl.	Aluminium, Sodium, Calcium, Oxygen (at STP), Copper chloride ($CuCl_2$).	Iron, Cobalt, Nickel, Gadolinium, Alnico, Lodestone.

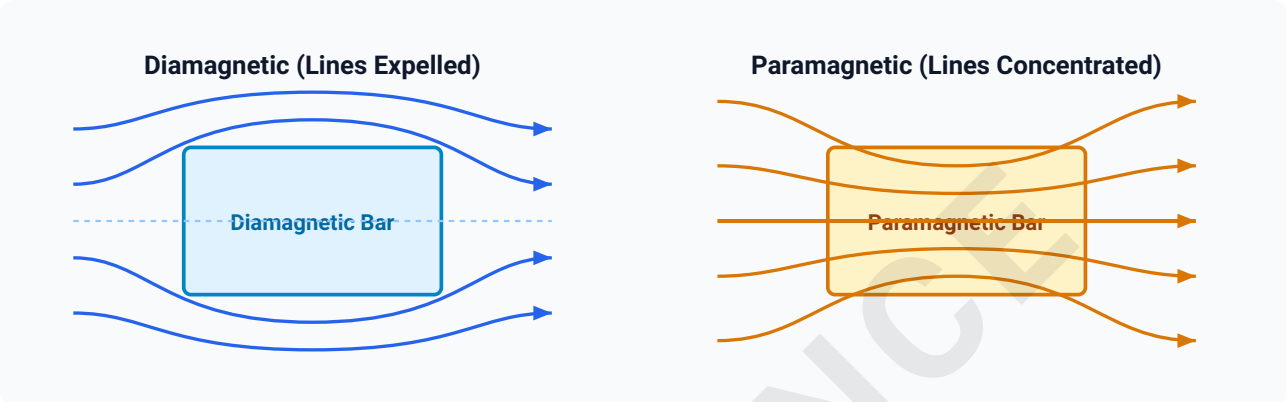


Figure 5: Modification of magnetic field lines when passing through (a) diamagnetic substance (repulsion/expulsion), and (b) paramagnetic substance (concentration/enhancement).

8.1 Diamagnetism and Lenz's Law Origin

In diamagnetic atoms, the orbital and spin magnetic moments of paired electrons cancel each other, leaving zero net ground-state atomic magnetic moment. When an external field $\mathbf{B}$ is applied:

- By **Faraday-Lenz's law of electromagnetic induction**, an electric field is induced on the orbiting electron.
- Electrons whose orbital magnetic moments are parallel to $\mathbf{B}$ slow down, while electrons with orbital moments antiparallel to $\mathbf{B}$ speed up.
- This induces a net macroscopic magnetic moment pointing in the direction *opposite* to the applied field $\mathbf{B}$, causing the material to be repelled from regions of stronger field to weaker field.

> **Universal Nature of Diamagnetism:** Diamagnetism is present in *all materials* as a fundamental relativistic and inductive response of orbiting electrons. However, in paramagnetic and ferromagnetic materials, the strong alignment of permanent dipoles overwhelms the weak diamagnetic effect, making it observable only in substances with zero permanent net atomic dipole moments.

8.2 Superconductors and the Meissner Effect

When specific metals and alloys are cooled below their critical superconducting transition temperature, they exhibit two simultaneous properties: **zero electrical resistance** (perfect conductivity) and **complete expulsion of internal magnetic flux** (perfect diamagnetism).

$$\chi = -1 \quad \Longrightarrow \quad \mu_r = 1 + \chi = 0 \quad \Longrightarrow \quad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = 0$$

This total magnetic flux expulsion is known as the **Meissner effect**. A superconductor forcefully repels a magnet (and by Newton's third law, is levitated above it), providing the physical mechanism behind high-speed magnetic levitation (Maglev) trains.

8.3 Ferromagnetism: Domains and Classification

In ferromagnetic materials, quantum-mechanical exchange interactions cause permanent atomic dipoles to spontaneously align parallel to one another over microscopic macroscopic volumes termed **domains**.

- Each unmagnetized domain is typically $\sim 1\,\text{mm}$ in linear dimension and contains roughly 10^{11} aligned atoms.

BOSONANCE

- In an unmagnetized bulk specimen, domain magnetization vectors are randomly oriented, resulting in zero net bulk magnetization.
- When an external field $\mathbf{B}_0$ is applied, two concurrent processes occur:
  1. Domains already aligned favorably with $\mathbf{B}_0$ *grow in volume* by shifting domain boundaries.
  2. Other domains rotate their magnetization vectors to align with $\mathbf{B}_0$, coalescing into a single giant domain.

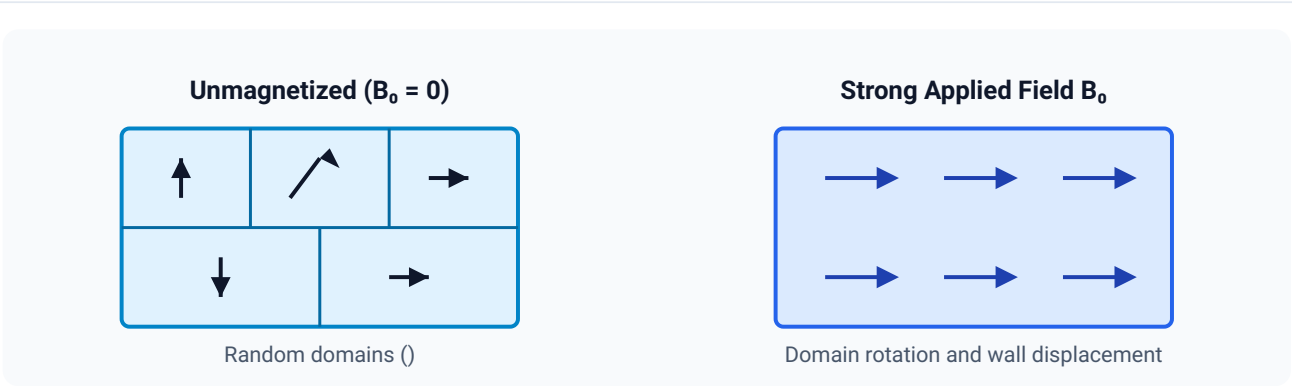


Figure 6: Domain structure of a ferromagnetic material: (a) Randomly oriented unmagnetized domains, and (b) Cooperative saturation alignment under an external magnetic field.

Hard vs. Soft Magnetic Materials:

- **Hard Ferromagnets:** Materials in which domain alignment and magnetization persist long after the external field is removed. They possess high retentivity and high coercivity. Examples: *Alnico* (an alloy of iron, aluminium, nickel, cobalt, and copper) and natural *lodestone*. Used to make permanent magnets.
- **Soft Ferromagnets:** Materials whose magnetization readily collapses upon removal of the applied field. They possess low coercivity. Example: *Soft iron*. Used in transformer cores, electromagnets, and telephone diaphragms.

9. Comprehensive Summary of Magnetic Quantities

Physical Quantity	Symbol	Vector / Scalar	Dimensional Formula	SI Unit	Defining Governing Relation
Permeability of Free Space	μ_0	Scalar	$[\mathrm{M\,L\,T^{-2}\,A^{-2}}]$	$\mathrm{T \cdot m \cdot A^{-1}}$ or $\mathrm{N \cdot A^{-2}}$	$\mu_0/4\pi = 10^{-7}\,\mathrm{T \cdot m \cdot A^{-1}}$
Magnetic Induction (B-Field)	$\mathbf{B}$	Vector	$[\mathrm{M\,T^{-2}\,A^{-1}}]$	Tesla (T) [$1\,\mathrm{T} = 10^4\,\mathrm{G}$]	$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$
Magnetic Dipole Moment	$\mathbf{m}$	Vector	$[\mathrm{L^2\,A}]$	$\mathrm{A \cdot m^2}$ or $\mathrm{J \cdot T^{-1}}$	$\mathbf{m} = NI\mathbf{A}$
Magnetic Flux	Φ_B	Scalar	$[\mathrm{M\,L^2\,T^{-2}\,A^{-1}}]$	Weber (Wb) [$1\,\mathrm{Wb} = 1\,\mathrm{T \cdot m^2}$]	$\Phi_B = \int \mathbf{B} \cdot d\mathbf{S}$
Magnetization	$\mathbf{M}$	Vector	$[\mathrm{L^{-1}\,A}]$	$\mathrm{A \cdot m^{-1}}$	$\mathbf{M} = \mathbf{m}_{\mathrm{net}}/V$
Magnetic Intensity	$\mathbf{H}$	Vector	$[\mathrm{L^{-1}\,A}]$	$\mathrm{A \cdot m^{-1}}$	$\mathbf{H} = (\mathbf{B}/\mu_0) - \mathbf{M}$
Magnetic Susceptibility	χ	Scalar	Dimensionless $[1]$	Unitless	$\mathbf{M} = \chi\mathbf{H}$
Relative Permeability	μ_r	Scalar	Dimensionless $[1]$	Unitless	$\mu_r = 1 + \chi = \mu/\mu_0$
Magnetic Permeability	μ	Scalar	$[\mathrm{M\,L\,T^{-2}\,A^{-2}}]$	$\mathrm{T \cdot m \cdot A^{-1}}$	$\mathbf{B} = \mu\mathbf{H}$

BOSONANCE

10. Fully Worked High-Yield Examples

Worked Example 1: Torque and Dipole Moment

NEET Core Standard

A short bar magnet placed with its axis at an angle of 30° with a uniform external magnetic field of $0.25\,\mathrm{T}$ experiences a torque of magnitude $4.5 \times 10^{-2}\,\mathrm{J}$.
(a) Calculate the magnitude of the magnetic moment of the magnet.
(b) Determine the work done by an external torque to rotate it from its most stable orientation to its most unstable orientation.

Step-by-Step Solution:
(a) The torque acting on a magnetic dipole in a uniform field is given by:

$$\tau = mB\sin\theta$$

Given: $\tau = 4.5 \times 10^{-2}\,\mathrm{J}$, $B = 0.25\,\mathrm{T}$, and $\theta = 30^\circ$ ($\sin 30^\circ = 0.5$).

$$4.5 \times 10^{-2} = m \times 0.25 \times 0.5 = 0.125\,m$$

$$m = \frac{4.5 \times 10^{-2}}{0.125} = \frac{4.5 \times 10^{-2}}{1/8} = 4.5 \times 10^{-2} \times 8 = 0.36\,\mathrm{J \cdot T^{-1}} \quad (\text{or } \mathrm{A \cdot m^2})$$

(b) The most stable position is $\theta_1 = 0^\circ$ ($U_1 = -mB$), and the most unstable position is $\theta_2 = 180^\circ$ ($U_2 = +mB$).

$$W = \Delta U = U_2 - U_1 = +mB - (-mB) = 2mB$$

$$W = 2 \times (0.36) \times (0.25) = 2 \times 0.09 = 0.18\,\mathrm{J}$$

Worked Example 2: Solenoid with a Magnetic Core

NEET Numerical Standard

A solenoid has 1000 turns per meter and carries a current of $2.0\,\mathrm{A}$. The core inside the solenoid is made of a ferromagnetic material with a relative permeability of $\mu_r = 400$.
Calculate: (a) the magnetic intensity H, (b) the total magnetic induction B, (c) the magnetization M, and (d) the equivalent magnetizing surface current I_m.

Step-by-Step Solution:
(a) The magnetic intensity H depends exclusively on the free conduction current:

$$H = nI = (1000\,\mathrm{m^{-1}}) \times (2.0\,\mathrm{A}) = 2.0 \times 10^3\,\mathrm{A \cdot m^{-1}}$$

(b) The total magnetic induction B in the core is:

$$B = \mu H = \mu_r \mu_0 H = 400 \times (4\pi \times 10^{-7}\,\mathrm{T \cdot m \cdot A^{-1}}) \times (2.0 \times 10^3\,\mathrm{A \cdot m^{-1}})$$

$$B = 400 \times 8\pi \times 10^{-4}\,\mathrm{T} = 3200\pi \times 10^{-4}\,\mathrm{T} = 0.32\pi\,\mathrm{T} \approx 1.005\,\mathrm{T} \approx 1.0\,\mathrm{T}$$

(c) The magnetization M is given by:

$$M = \chi H = (\mu_r - 1)H = (400 - 1) \times (2.0 \times 10^3) = 399 \times 2.0 \times 10^3 = 7.98 \times 10^5\,\mathrm{A \cdot m^{-1}} \approx 8.0 \times 10^5\,\mathrm{A \cdot m^{-1}}$$

(d) The magnetizing current I_m is the additional conduction current needed in the air-core solenoid to produce the same B field of $1.0\,\mathrm{T}$:

$$B = \mu_0 n(I + I_m) \implies I + I_m = \frac{B}{\mu_0 n} = \frac{1.0}{(4\pi \times 10^{-7}) \times 1000} \approx \frac{1.0}{1.257 \times 10^{-3}} \approx 796\,\mathrm{A}$$

$$I_m = 796 - 2 = 794\,\mathrm{A}$$

BOSONANCE

Worked Example 3: Axial and Equatorial Magnetic Field

Formula Application

A short bar magnet has a magnetic moment of $0.48\,\mathrm{J\cdot T^{-1}}$. Find the magnitude and direction of the magnetic field produced by the magnet at a distance of $10\,\mathrm{cm}$ $(0.1\,\mathrm{m})$ from its center on:
(a) its axial line,
(b) its equatorial line.

Step-by-Step Solution:
(a) For a short bar magnet, the field along the axis is:

$$B_A = \frac{\mu_0}{4\pi}\frac{2m}{r^3}$$

Given: $\mu_0/4\pi = 10^{-7}\,\mathrm{T\cdot m\cdot A^{-1}}$, $m = 0.48\,\mathrm{J\cdot T^{-1}}$, $r = 0.1\,\mathrm{m} \implies r^3 = 10^{-3}\,\mathrm{m^3}$.

$$B_A = 10^{-7} \times \frac{2 \times 0.48}{10^{-3}} = 10^{-7} \times \frac{0.96}{10^{-3}} = 0.96 \times 10^{-4}\,\mathrm{T} = 0.96\,\mathrm{G}$$

Direction: Directed along the direction of the magnetic dipole moment vector $\mathbf{m}$ (from S to N).

(b) The field along the equatorial line (normal bisector) is:

$$B_E = \frac{\mu_0}{4\pi}\frac{m}{r^3} = \frac{B_A}{2} = \frac{0.96 \times 10^{-4}}{2} = 0.48 \times 10^{-4}\,\mathrm{T} = 0.48\,\mathrm{G}$$

Direction: Directed strictly opposite (antiparallel) to the magnetic dipole moment vector $\mathbf{m}$.

Worked Example 4: Cutting Magnets

Conceptual Analysis

A uniform bar magnet of magnetic moment M and length L is cut into two equal halves. Find the new magnetic moment in each piece if the cut is made:
(a) transverse (perpendicular) to its length,
(b) longitudinal (parallel) along its length.

Step-by-Step Solution:
Recall that magnetic dipole moment is defined as the product of pole strength q_m and magnetic length L: $M = q_m \times L$.

(a) **Transverse Cut:** The cross-sectional area remains unchanged, so the pole strength q_m remains the same. However, the length of each half becomes $L' = L/2$.

$$M' = q_m \times L' = q_m \times \frac{L}{2} = \frac{M}{2}$$

(b) **Longitudinal Cut:** The length remains L, but the cross-sectional area of each half is halved, so the pole strength is halved ($q_m' = q_m/2$).

$$M' = q_m' \times L = \frac{q_m}{2} \times L = \frac{M}{2}$$

Conclusion: In both cases, the magnetic moment of each resulting piece is exactly $M/2$.

BOSONANCE

11. Original NEET UG Exam-Style Practice Questions

Each practice question below is designed in strict adherence to current NEET UG formats, covering calculation-based MCQs, assertion-reason items, multi-statement evaluations, and matching tables with complete step-by-step reasoning.

Practice Question 1

Calculation MCQ

A closely wound solenoid of 800 turns and cross-sectional area $2.5 \times 10^{-4}\,\mathrm{m}^2$ carries a steady current of $3.0\,\mathrm{A}$. When suspended horizontally in a uniform horizontal magnetic field of $0.25\,\mathrm{T}$, what is the magnitude of the torque experienced when the axis of the solenoid makes an angle of 30° with the magnetic field lines?

(A) $0.0375\,\mathrm{N \cdot m}$

(B) $0.075\,\mathrm{N \cdot m}$

(C) $0.15\,\mathrm{N \cdot m}$

(D) $0.60\,\mathrm{N \cdot m}$

Correct Answer: (B) $0.075\,\mathrm{N \cdot m}$
Explanation:
1. First, calculate the magnetic dipole moment m of the solenoid:

$$m = NIA = 800 \times 3.0\,\mathrm{A} \times (2.5 \times 10^{-4}\,\mathrm{m}^2) = 2400 \times 2.5 \times 10^{-4} = 0.6\,\mathrm{A \cdot m}^2 \quad (\text{or } \mathrm{J \cdot T^{-1}})$$

2. The torque on the solenoid is:

$$\tau = mB\sin\theta = 0.6 \times 0.25 \times \sin 30^\circ = 0.15 \times 0.5 = 0.075\,\mathrm{N \cdot m}$$

Practice Question 2

Assertion & Reason

Given below are two statements labeled as Assertion (A) and Reason (R):
Assertion (A): The net magnetic flux emerging through any arbitrary closed Gaussian surface is identically zero.
Reason (R): Isolated magnetic monopoles do not exist, and magnetic field lines form continuous closed loops without sources or sinks.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)
Explanation:
Gauss's law for magnetism states that $\oint \mathbf{B} \cdot d\mathbf{S} = 0$. This mathematical identity directly stems from the empirical fact that magnetic field lines never begin or terminate at a point; every line entering a closed surface must leave it. This signifies the absence of isolated magnetic charges (monopoles). Hence, both Assertion and Reason are true, and the Reason correctly explains the Assertion.

BOSONANCE

Practice Question 3

Statement-Based MCQ

Consider the following statements regarding magnetic properties of materials:
Statement I: A superconductor is a perfect diamagnetic substance with magnetic susceptibility $\chi = -1$ and relative permeability $\mu_r = 0$.
Statement II: The magnetic susceptibility of a diamagnetic substance increases rapidly with an increase in absolute temperature.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (C)
Explanation:
Statement I is correct: Under the Meissner effect, superconductors completely expel internal magnetic flux, leading to $\mathbf{B} = 0$, which yields $\chi = -1$ and $\mu_r = 1 + \chi = 0$.
Statement II is incorrect: Diamagnetism arises from induced orbital electronic currents according to Lenz's law and is essentially *independent of temperature*. Paramagnetic susceptibility obeys Curie's law ($\chi \propto 1/T$), not diamagnetic susceptibility.

Practice Question 4

Matching List

Match the physical substances in **List I** with their corresponding magnetic classifications in **List II**:

List I (Substance)	List II (Magnetic Behavior)
(P) Bismuth	(1) Ferromagnetic with $\mu_r > 1000$
(Q) Aluminium	(2) Perfect diamagnet with $\chi = -1$
(R) Alnico	(3) Diamagnetic with $\chi \approx -10^{-5}$
(S) Superconductor	(4) Paramagnetic with $\chi \approx +10^{-5}$

(A) P-3, Q-4, R-1, S-2

(B) P-4, Q-3, R-1, S-2

(C) P-3, Q-1, R-4, S-2

(D) P-2, Q-4, R-1, S-3

Correct Answer: (A) P-3, Q-4, R-1, S-2
Explanation:
• Bismuth is a classic diamagnetic material ($\chi \approx -10^{-5}$).
• Aluminium is a standard paramagnetic material ($\chi \approx +10^{-5}$).
• Alnico (alloy of Al, Ni, Co, Fe, Cu) is a hard ferromagnetic material ($\mu_r \gg 1$).
• A superconductor exhibits the Meissner effect ($\chi = -1$).

BOSONANCE

Practice Question 5

Work Done on Dipole

A magnetic dipole of magnetic moment $m = 1.5\,\text{J}\cdot\text{T}^{-1}$ is initially aligned parallel to a uniform magnetic field of $B = 0.22\,\text{T}$. What is the external work required to rotate the dipole by 90° and by 180° from its initial position?

(A) $0.165\,\text{J}$ and $0.33\,\text{J}$

(B) $0.33\,\text{J}$ and $0.66\,\text{J}$

(C) $0.33\,\text{J}$ and $0\,\text{J}$

(D) $0.66\,\text{J}$ and $1.32\,\text{J}$

Correct Answer: (B) $0.33\,\text{J}$ and $0.66\,\text{J}$

Explanation:

Work required by external agent is $W = mB(\cos\theta_1 - \cos\theta_2)$, with $\theta_1 = 0^\circ$.

1. For rotation to $\theta_2 = 90^\circ$:

$$W_1 = mB(\cos 0^\circ - \cos 90^\circ) = mB(1 - 0) = mB$$
$$W_1 = 1.5\,\text{J}\cdot\text{T}^{-1} \times 0.22\,\text{T} = 0.33\,\text{J}$$

2. For rotation to $\theta_2 = 180^\circ$:

$$W_2 = mB(\cos 0^\circ - \cos 180^\circ) = mB(1 - (-1)) = 2mB$$
$$W_2 = 2 \times 0.33\,\text{J} = 0.66\,\text{J}$$

Practice Question 6

Conceptual Comparison

Two identical looking bars of iron, labeled X and Y, are given. One of them is a permanent magnet, while the other is an unmagnetized soft iron bar. Without using any third object or apparatus, how can you unambiguously determine which bar is the magnet?

(A) Touch one end of X to one end of Y; if they attract, both are magnets.

(B) Touch the end of X to the center of Y; if there is attraction, Y is the magnet.

(C) Touch the end of X to the center of Y; if there is no attraction, Y is the magnet.

(D) Suspend both bars horizontally; the unmagnetized bar will spin continuously.

Correct Answer: (C)

Explanation:

In a bar magnet, the magnetic field intensity is strongest at the two ends (poles) and practically zero at the geometric middle.

• If you touch an end of bar X to the middle of bar Y, and observe *no attractive force*, it proves that the middle of bar Y has no magnetic poles; hence **bar Y is the magnet** (and bar X is merely an unmagnetized iron bar attracted to Y's poles, but finding no field at Y's center).

• Conversely, if X were the magnet, its pole tip would attract the middle of unmagnetized iron bar Y with full strength. Thus, option (C) is completely correct.

12. Common NEET Pitfalls & Revision Checklist

Five Critical Pitfalls to Avoid in NEET UG:

1. **Confusing $\mathbf{B}$ and $\mathbf{H}$:** Remember that $\mathbf{H}$ is determined solely by external free currents ($H = nI$ in a solenoid), whereas $\mathbf{B}$ accounts for the medium's magnetization as well ($\mathbf{B} = \mu\mathbf{H}$). Both $\mathbf{H}$ and $\mathbf{M}$ share identical units of $\text{A}\cdot\text{m}^{-1}$, while $\mathbf{B}$ is measured in Tesla (T).
2. **The Minus Sign in Equatorial Field:** Do not omit the vector direction! The axial field $\mathbf{B}_A$ points *parallel* to the dipole moment $\mathbf{m}$, but the equatorial field $\mathbf{B}_E$ points *antiparallel* ($-\mathbf{m}$). The ratio of their magnitudes is strictly $B_A/B_E = 2$ at identical large distances.

3. **Zero Potential Energy vs. Equilibrium:** A dipole at $\theta = 90^\circ$ has $U = 0$, but experiences *maximum torque* ($\tau = mB$); it is NOT in equilibrium. Equilibrium requires $\tau = 0$, which occurs at $\theta = 0^\circ$ (stable, $U = -mB$) and $\theta = 180^\circ$ (unstable, $U = +mB$).
4. **Flux through Open vs. Closed Surfaces:** Gauss's law for magnetism states that flux through a *closed* surface is zero ($\oint \mathbf{B} \cdot d\mathbf{S} = 0$). The magnetic flux through an *open* surface (such as a single coil loop) is generally non-zero ($\Phi = \mathbf{B} \cdot \mathbf{A}$).
5. **Susceptibility Signs:** Diamagnets have negative susceptibility ($-1 \leq \chi < 0$), yielding $\mu_r < 1$. Paramagnets have small positive susceptibility ($0 < \chi < 10^{-4}$), giving $\mu_r > 1$. Ferromagnets have very large positive susceptibility ($\chi \gg 1$), giving $\mu_r \gg 1$.

Quick NEET Revision Checklist

- ☑ Magnetic dipole moment formula: $\mathbf{m} = NI\mathbf{A}$ (Units: $\mathrm{A} \cdot \mathrm{m}^2$ or $\mathrm{J} \cdot \mathrm{T}^{-1}$).
- ☑ Torque on dipole: $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$; Potential Energy: $U = -\mathbf{m} \cdot \mathbf{B}$.
- ☑ Work done in rotating dipole: $W = mB(\cos\theta_1 - \cos\theta_2)$.
- ☑ Short dipole axial field: $B_A = \frac{\mu_0}{4\pi}\frac{2m}{r^3}$; equatorial field: $B_E = \frac{\mu_0}{4\pi}\frac{m}{r^3}$.
- ☑ Permeability relations: $\mu_r = 1 + \chi$, $\mu = \mu_0\mu_r = \mu_0(1+\chi)$.
- ☑ Superconductors: Perfect diamagnets, $\chi = -1$, $\mu_r = 0$, $\mathbf{B} = 0$ (Meissner effect).
- ☑ Gauss's law: $\oint \mathbf{B} \cdot d\mathbf{S} = 0$ (No magnetic monopoles).

13. Concise Final Exam Recap

Magnetism originates microscopically from circulating electronic currents and atomic spins. Magnetic field lines are continuous closed loops entering the South pole and exiting the North pole, ensuring that the net magnetic flux through any closed surface is identically zero ($\oint \mathbf{B} \cdot d\mathbf{S} = 0$). A bar magnet behaves identically to a finite current-carrying solenoid with dipole moment $m = NIA$. Inside materials, the total field is given by $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$. In response to external magnetic fields, materials are classified into:

- **Diamagnetic:** Universal, weak repulsion from strong to weak field, $\chi < 0$, temperature-independent, perfect in superconductors ($\chi = -1$).
- **Paramagnetic:** Weak attraction from weak to strong field, $\chi > 0$ (small), permanent atomic dipoles, obeys Curie's law ($\chi \propto 1/T$).
- **Ferromagnetic:** Strong attraction, domain cooperative alignment, $\chi \gg 1$, $\mu_r \gg 1$, classified into hard ferromagnets (permanent magnets, high coercivity like Alnico) and soft ferromagnets (transformer cores, low coercivity like soft iron).

BOSONANCE